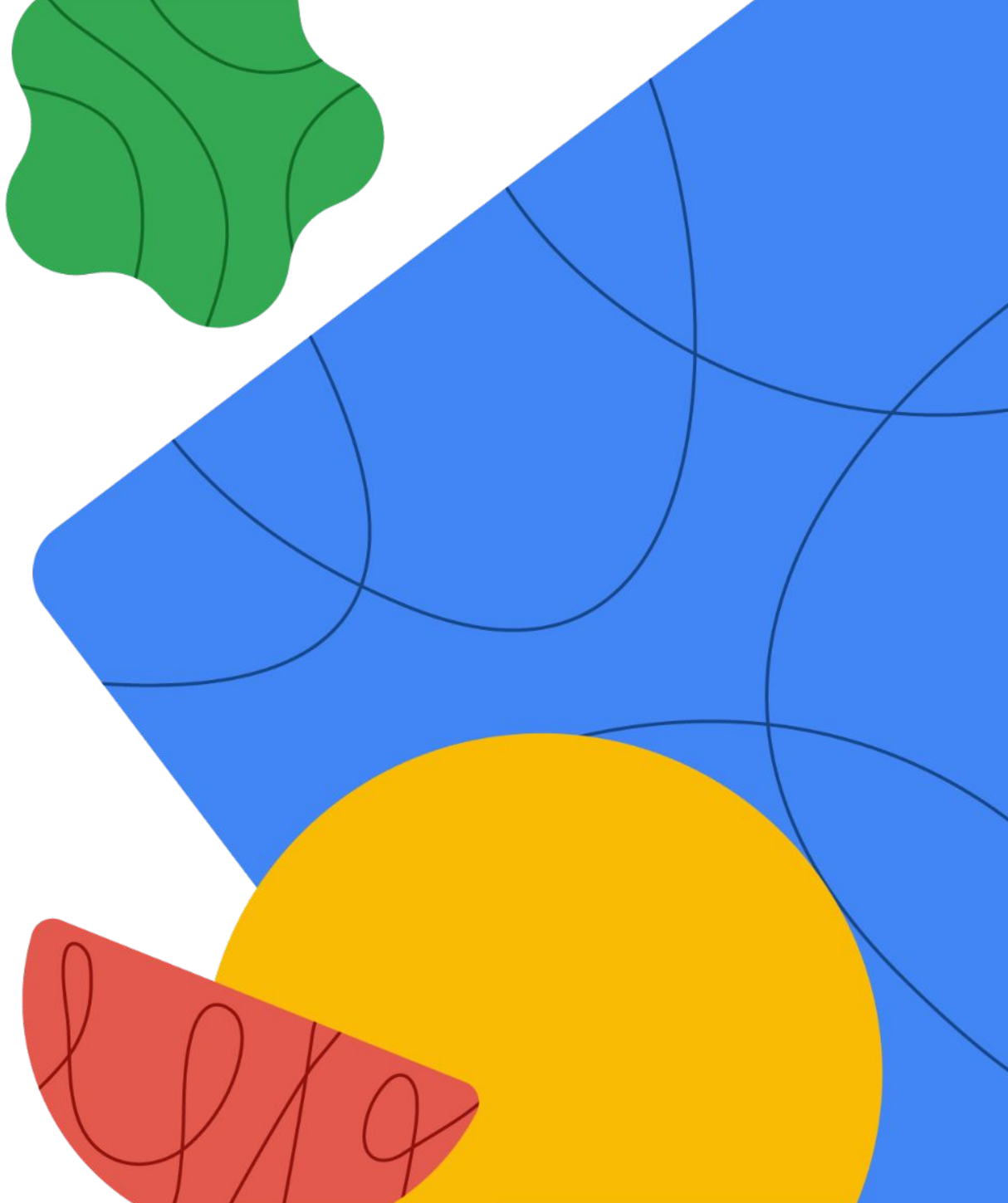
Google Cloud

Introduction to AI and Machine Learning on Google Cloud

Module 1
AI foundations



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01 AI foundations

02 Generative AI

03 AI development options

04 AI development workflow

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Predictive AI versus generative AI

Predictive AI

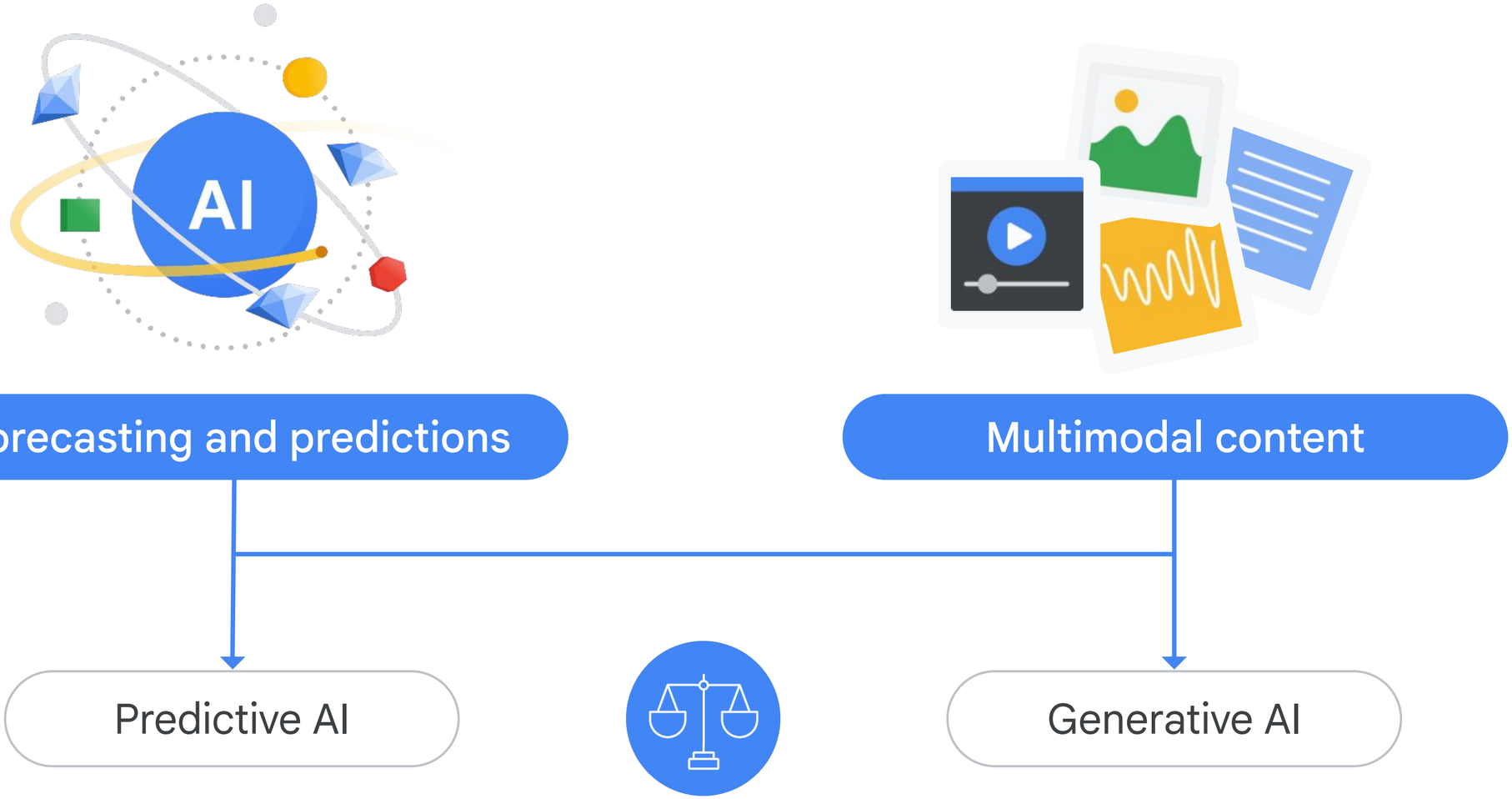
- Uses existing data to classify information.
- Predicts future outcomes based on historical patterns.
- Excels at learning what's already there to make informed decisions.

Generative AI

- Creates summaries.
- Uncovers complex correlations.
- Generates new content such as, text, images, or videos.

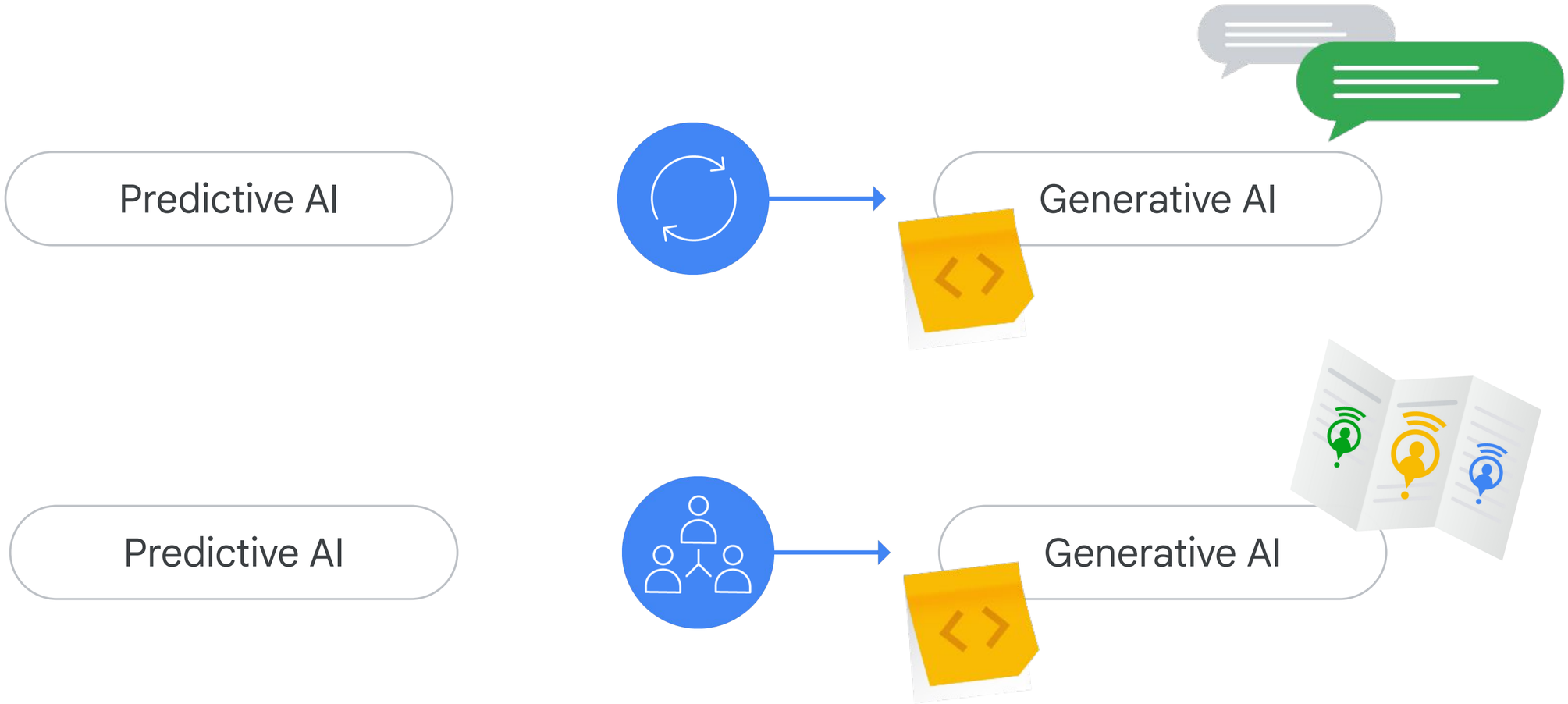
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Decision tree



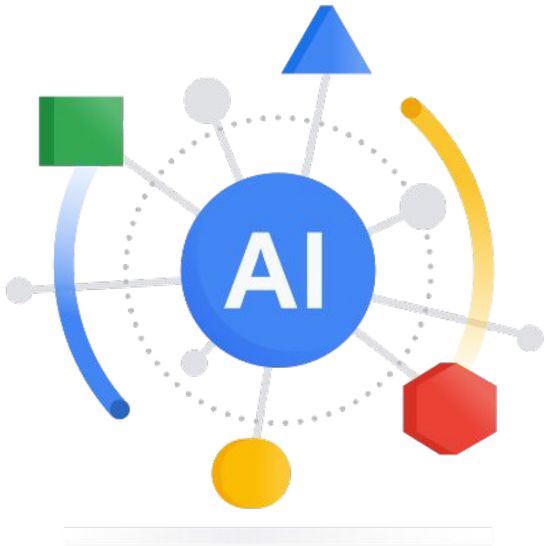
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Combine predictive AI and generative AI

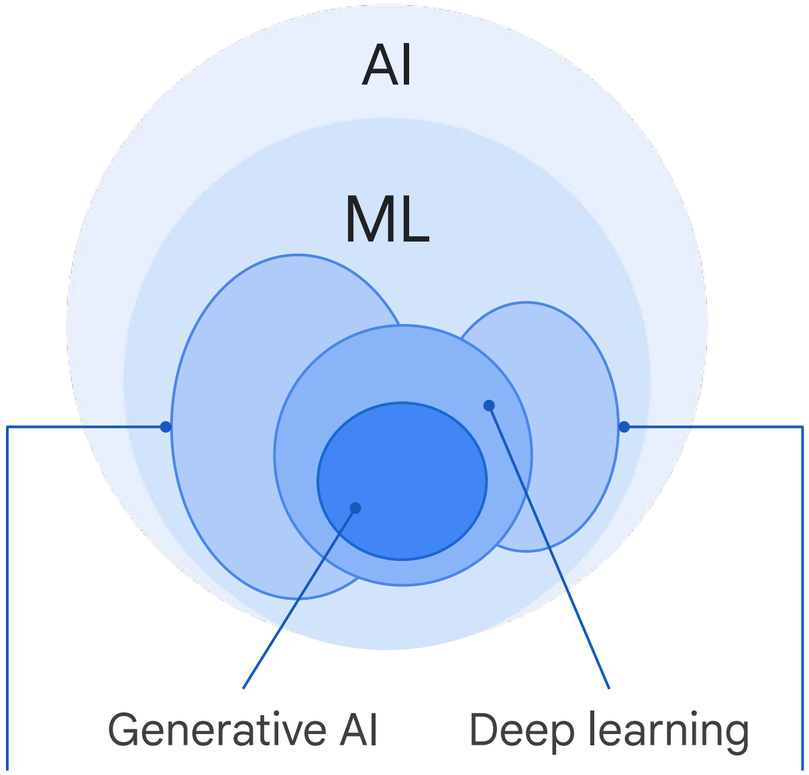


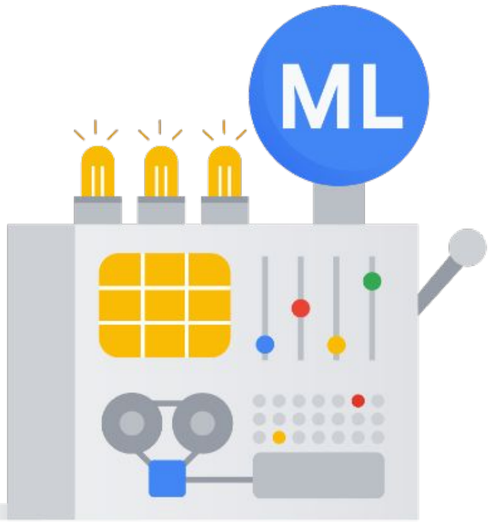
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Artificial intelligence versus machine learning



Artificial intelligence





Machine learning

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Supervised learning



Labeled data, task-driven, identifies a goal.

Classification

Predicts a categorical variable.

Determines whether a picture shows a cat or a dog.

Uses logistic regression.

Regression

Predicts a numeric variable.

Forecasts sales for a product based on its past sales.

Uses linear regression.

Unsupervised learning



Unlabeled data, data-driven, identifies a pattern.

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Supervised learning



Labeled data, task-driven, identifies a goal.

Unsupervised learning



Unlabeled data, data-driven, identifies a pattern.

Clustering

Groups data points together.

Uses customer demographics to determine customer segmentation.

Uses k-means clustering.

Association

Identifies underlying relationships.

Correlates two products to place them closer in a grocery store.

Uses association rule learning.

Dimensionality reduction

Reduces the number of dimensions.

Combines characteristics to create a simplified rule for an insurance quote.

Uses principal component analysis.

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Use case 1

Example 1

Predict customer spending based on purchase history.

Date	Customer ID	Age	Gender	Address	Amount
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x

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Did you get the answers right?

Example 1

Predict customer spending based on purchase history.

Date	Customer ID	Age	Gender	Address	Amount
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x

- Supervised learning
- Regression problem
- Linear regression

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Use case 2

Example 2

Identify customer segmentation.

Date	Customer ID	Age	Gender	Address	Amount
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x

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Did you get the answers right?

Example 2

Identify customer segmentation.

Date	Customer ID	Age	Gender	Address	Amount
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x

- Unsupervised learning
- Clustering problem
- Clustering analysis

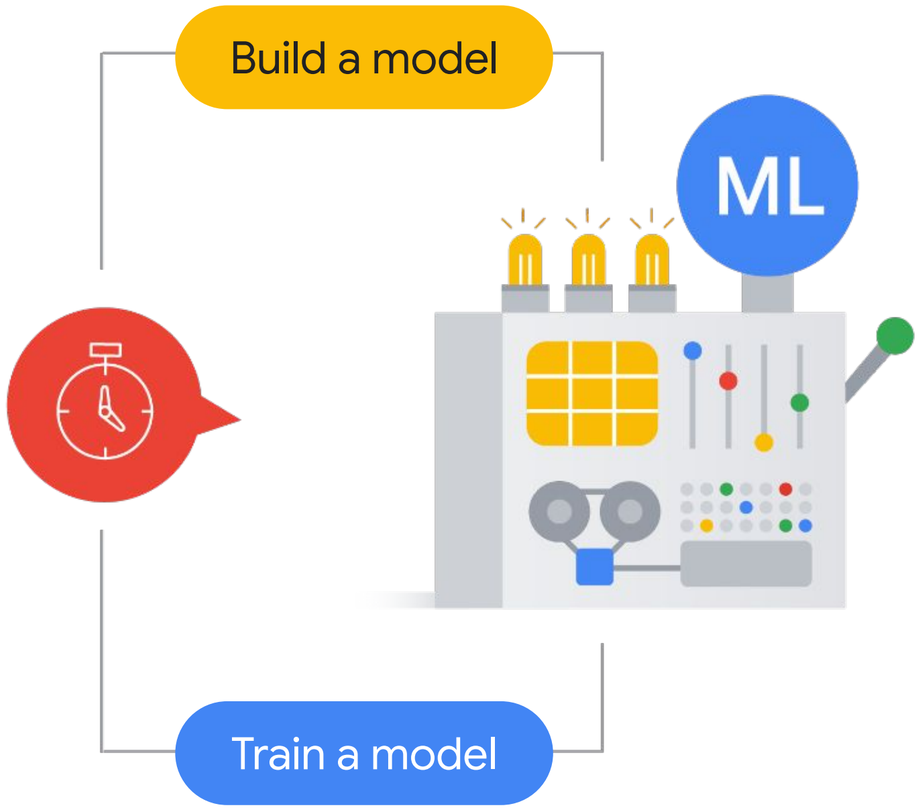
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BigQuery ML

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Challenges to build and train ML models



01 Import and prepare data.

02 Experiment with different models.

03 Train the model with new data.

04 Deploy the model to make predictions.